



BUSINESS ANALYSIS

Lecture 5: Requirements Engineering

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WHAT IS REQUIREMENTS ENGINEERING?

Definition

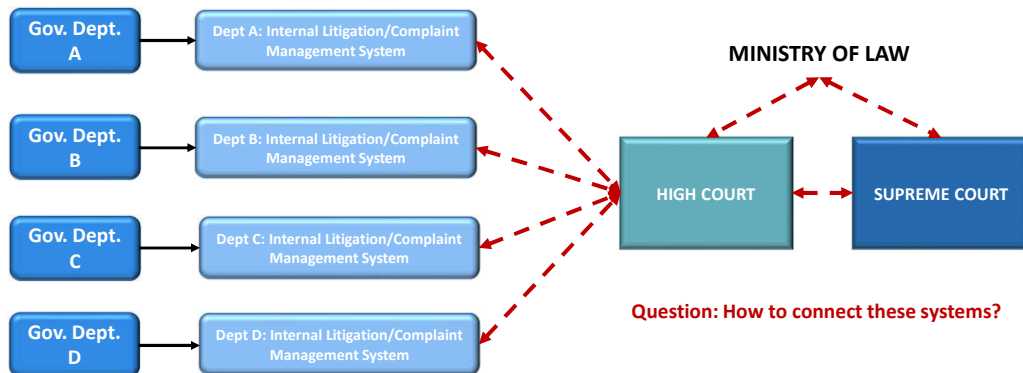
- "Requirements engineering is the initial step of software development activity in which the requirements from the customer are elicited and documented.....The typical definition of engineering refers to the creation of cost effective solution to practical problems by applying scientific knowledge. A process involved in developing the system requirements are called **requirements engineering**." (Shams-ul-Arif et. al., 2009)
- A software-intensive system is an interrelated set of human activities, supported by computer technology. The requirements express the purpose of such a system (modified from Steve Easterbrook, 2004)
- "Requirements engineering is widely concerned with the elicitation, analysis, specification, and validation of software requirements." (Yue-min Zhao & Junwen Zhao, 2018)

Source: See reference/reading list at the end of the presentation

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INTRODUCITON: REQUIREMENTS ENGINEERING

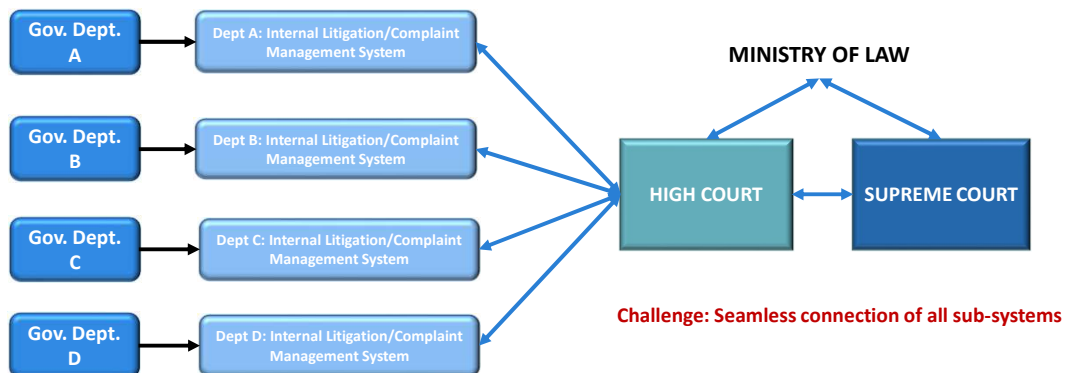
EXAMPLE: Litigation and Complaint Management Systems (LMS)



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INTRODUCITON: REQUIREMENTS ENGINEERING

EXAMPLE: Litigation and Complaint Management Systems (LMS)



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WHAT IS REQUIREMENTS ENGINEERING?

Rationale for requirements engineering

- Provides foundation for all the other activities of software development such as designing, implementation, testing, operation and maintenance
- According to one study errors made at the requirements stage account for 40% to 60% defects in software projects
- According to another study 13.1% projects fail due to incomplete requirements and 8.8% projects fail due to rapidly changing requirements (scope creep?)

Source: See reference/reading list at the end of the presentation

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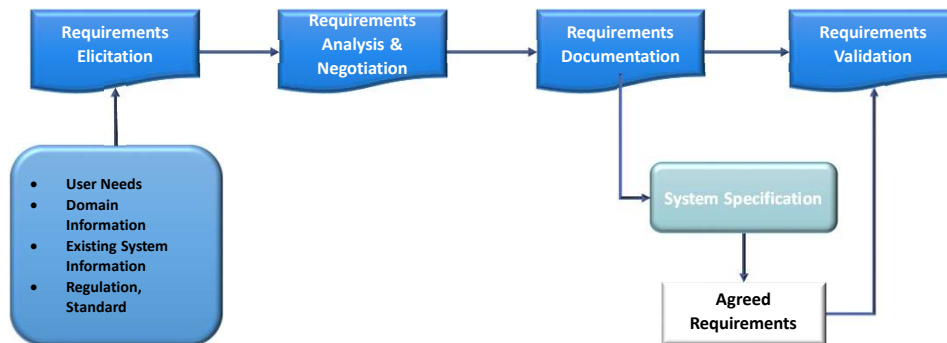
MORE ABOUT REQUIREMENTS ENGINEERING

“Computer systems are designed, and anything that is designed has an intended purpose. If a computer system is unsatisfactory, it is because the system was designed without an adequate understanding of its purpose, or because we are using it for a purpose different from the intended one. Requirements Engineering provides a framework for understanding the purpose of a system and the contexts in which it will be used. Or, put another way, requirements engineering bridges the gap between an initial vague recognition that there is some problem to which we can apply computer technology, and the task of building a system to address the problem.” (Easterbrooks, 2004)

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THE PROCESS OF REQUIREMENTS ENGINEERING

Linear Requirements Engineering Model



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IDENTIFYING AND MANAGING STAKEHOLDERS

- Who are stakeholders?

In the context of a “software project, stakeholders are individuals or groups who can affect or be affected by the achievement of project’s objectives” (Yuemin Zhao & Junwen Zhao, 2018)

- Stakeholder management

- Stakeholder identification
- Power-Interest matrix
- Communication planning
- Conflict management

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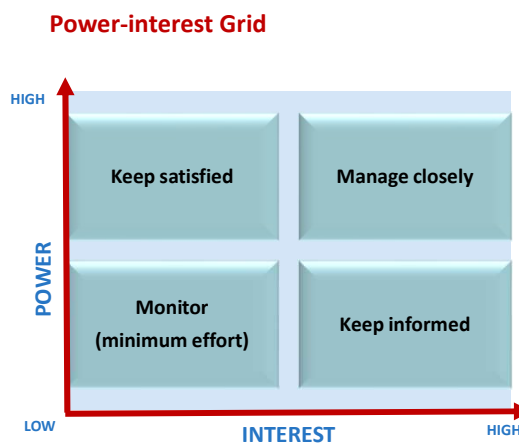
STAKEHOLDER IDENTIFICATION

- **Why is it important to identify stakeholders?**
 - For understanding purpose and expectations
 - As a pre-requisite to selection of important stakeholders so that time and cost for eliciting opinion of unimportant stakeholders can be saved
- **Stakeholder types**
 - Customers or clients: Those who pay for the system (e.g., donors, sponsors, business clients, investors)
 - Developers: Those who design, construct, and maintain the software
 - Users: People who interact with the system for a task or purpose

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STEPS IN STAKEHOLDER IDENTIFICATION

- Classification of stakeholders
 - critical stakeholders
 - major-stakeholders
 - normal-stakeholders
 - minor-stakeholders
- Prioritizing the stakeholders



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POWER-INTEREST GRID AND CONFLICT MANAGEMENT

Why is it important to map the stakeholders through the Power-Interest Grid important?

- Stakeholder groups can have conflicting interests and objectives, leading to tensions and conflicts
Example: Automation endeavors by management vis-à-vis staff interest in retention of jobs which may be lost to automation
- Challenge for business analyst: How to balance the competing demands of different stakeholder groups?
Example: CEO of a company wants to invest in technology to improve efficiency, but the junior staff may resist since it may reduce their discretionary power
- Conflicts can also arise when stakeholders have different levels of power and influence
Example: Senior management of a department may resist e-government projects since it may reduce their discretionary power
- Managing stakeholder conflicts requires careful communication, transparency, and compromise
- Example of conflict: Digitizing the land record system

Video link: <https://www.youtube.com/watch?v=PdZlkWfM1pl>

Question: How was the conflict managed?

Source: save my exams; *Stakeholder Conflicts*: Available at <https://www.savemyexams.com/dp/business/ib/management/24/hl/revision-notes/1-introduction-to-business-management/1-4-stakeholders/stakeholder-conflicts/>

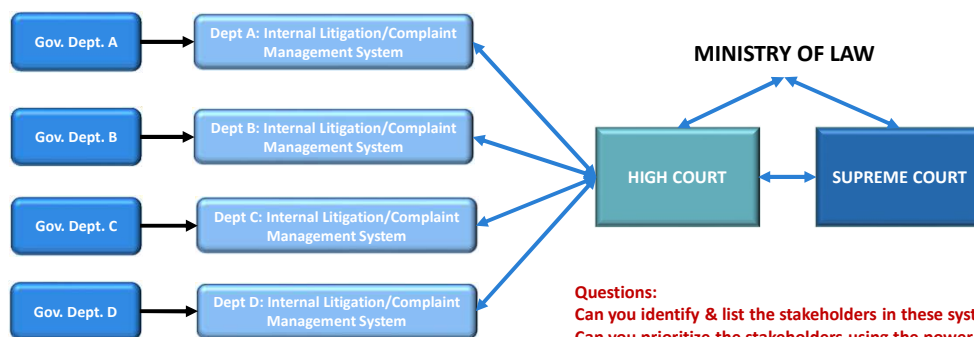
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TEN MINUTES ACTIVITY

EXAMPLE: Litigation and Complaint Management Systems (LMS)

Video: <https://web.facebook.com/watch/?v=1645868799400882>



Questions:

Can you identify & list the stakeholders in these systems?

Can you prioritize the stakeholders using the power-interest grid?

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MANAGING STAKEHOLDERS: COMMUNICATION PLAN

- What is a stakeholders communications plan?

“A stakeholder communications plan is a strategic document that how an organization will communicate with stakeholders”

- Why create a communications plan?

- Coordinate
- Collaborate
- Plan

Source: simply stakeholders; *How to Create a Stakeholder Communications Plan*: Available at <https://simplystakeholders.com/stakeholder-communications-plan/>

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MANAGING STAKEHOLDERS: COMMUNICATION PLAN

Steps in creating a stakeholders communication plan

- Analyze the situation: Understand who you are working with and why
- Define your goals: What are the desired outcomes or deliverables
- Describe your audience: Who is at the receiving end of your communication
- Get tactical: What messages, communication platforms and channels should you use
- Assign roles & responsibilities: Who is responsible for the various tasks in your plan
- Map your timelines: What is the schedule of your communication plan
- Monitor and evaluate: How are you going to monitor and evaluate your communication plan

Source: simply stakeholders; *How to Create a Stakeholder Communications Plan*: Available at <https://simplystakeholders.com/stakeholder-communications-plan/>

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ESSENTIAL READINGS

Project Management.com, "Stakeholder Analysis using the Power Interest Grid", available at:
<https://www.projectmanagement.com/wikis/368897/stakeholder-analysis--using-the-power-interest-grid>

save my exams; *Stakeholder Conflicts*: Available at <https://www.savemyexams.com/dp/business/ib/management/24/hl/revision-notes/1-introduction-to-business-management/1-4-stakeholders/stakeholder-conflicts/>

Shams-ul-arif, Mr. Qadeem Khan, S. A. K. Gahyyur, 2009-2010, International Journal of Reviews in Computing, "Requirements Engineering Processes, Tools/Technologies, & Methodologies."

simply stakeholders; *How to Create a Stakeholder Communications Plan*: Available at <https://simplystakeholders.com/stakeholder-communications-plan/>

Steve Easterbrooks, 2004, "What is requirements engineering?"

Yuemin Zhao & Junwen Zhao, 2018, A Systematic Stakeholder Selection Model in Requirements Elicitation for Software Projects: A Systematic Mapping Study

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REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK)**; IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)

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